



Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and product preferences from
3,900 transactions

Dataset at a Glance

3,900

Purchases

37

Missing Values

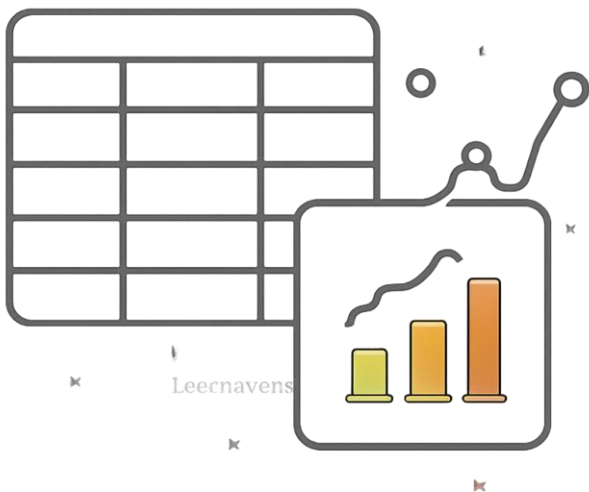
Review Rating only

18

Columns

\$59.76

Avg. Purchase



Key Features

- Demographics: Age, Gender, Location, Subscription
- Purchases: Item, Category, Amount, Season
- Behavior: Discount, Promo Code, Frequency, Rating



Data Preparation in Python

01

Load & Explore

Imported via pandas; `df.info()` and `describe()`

02

Clean & Standardize

Imputed missing Review Ratings by category median; renamed to snake_case

03

Feature Engineering

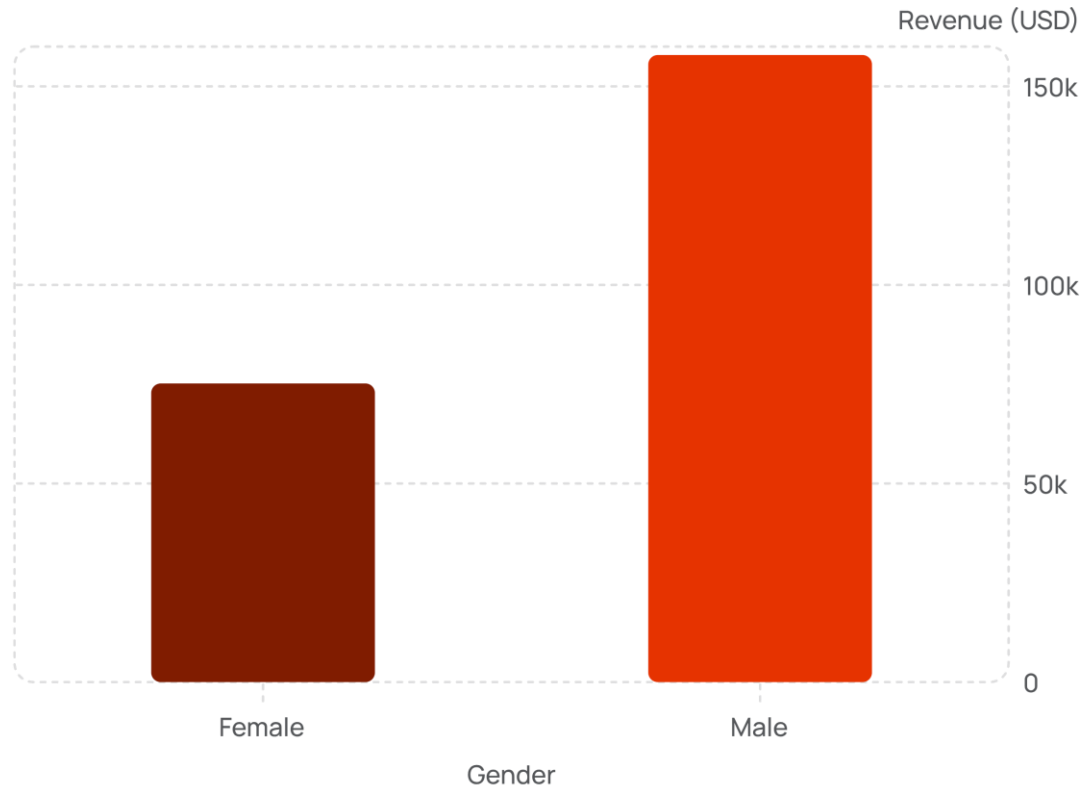
Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`

04

Database Load

Connected to PostgreSQL; loaded cleaned DataFrame for SQL analysis

Revenue by Gender & Subscription



Male customers generate **2× more revenue**

Subscribers vs. Non-Subscribers

Subscribers (27%)

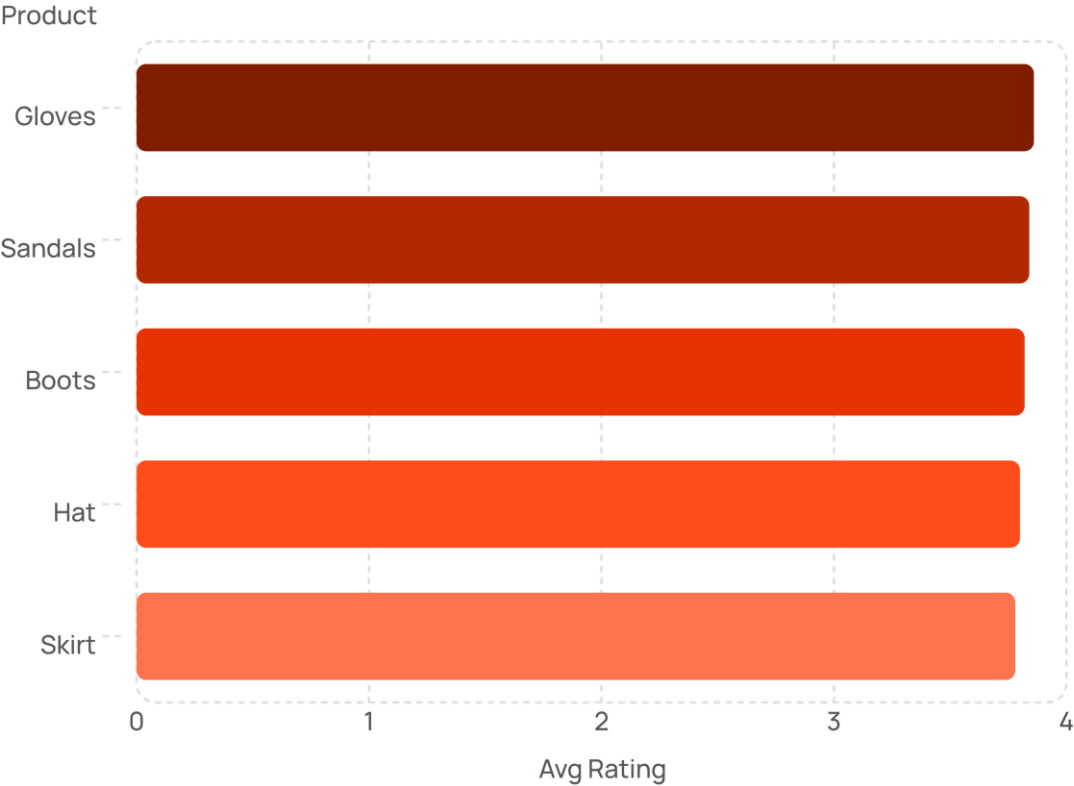
1,053 customers ·
\$62,645 total

Non-Subscribers (73%)

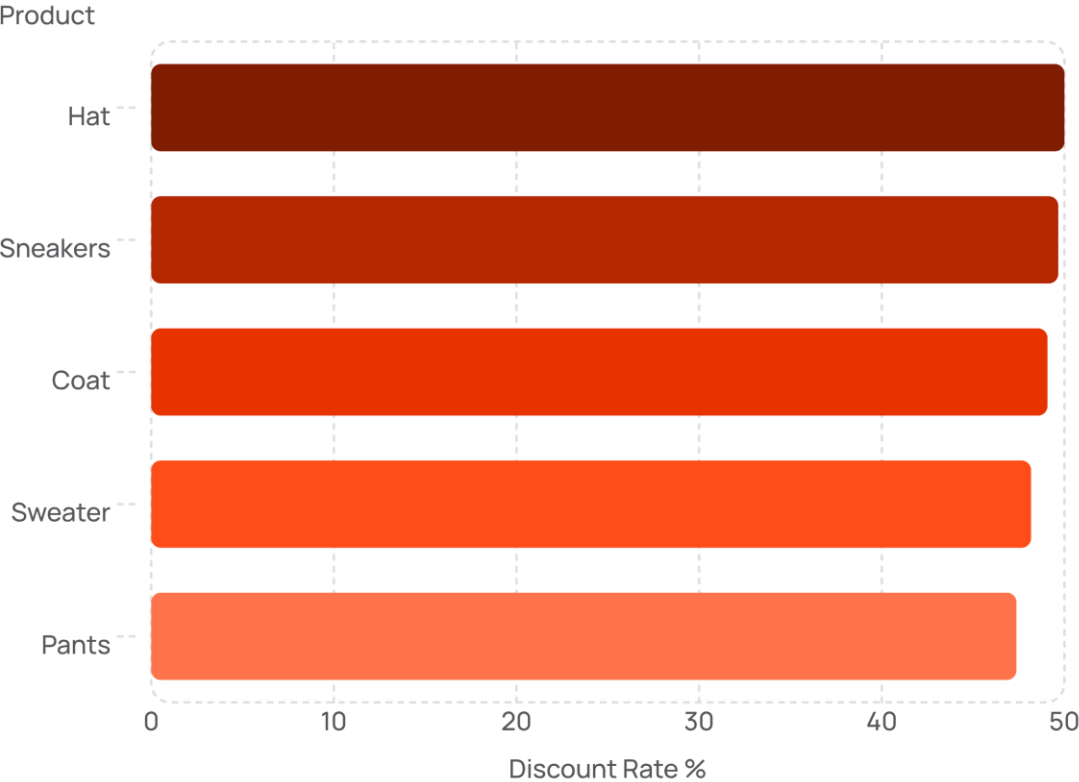
2,847 customers ·
\$170,436 total

Top Products & Discount Patterns

Top 5 by Review Rating



Most Discounted Products



Customer Segmentation

3,116

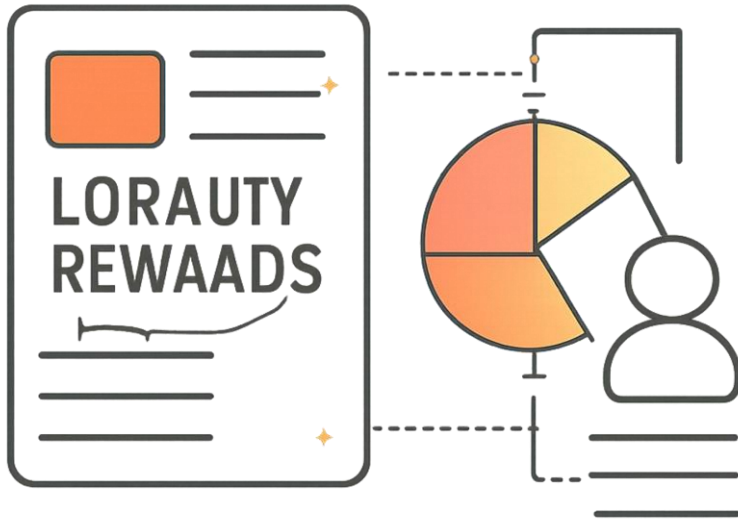
Loyal

701

Returning

83

New



839 high-spending discount users spent above average purchase amount despite using discounts.

Top Products by Category



Accessories

Jewelry · Sunglasses · Belt



Clothing

Blouse · Pants · Shirt



Footwear

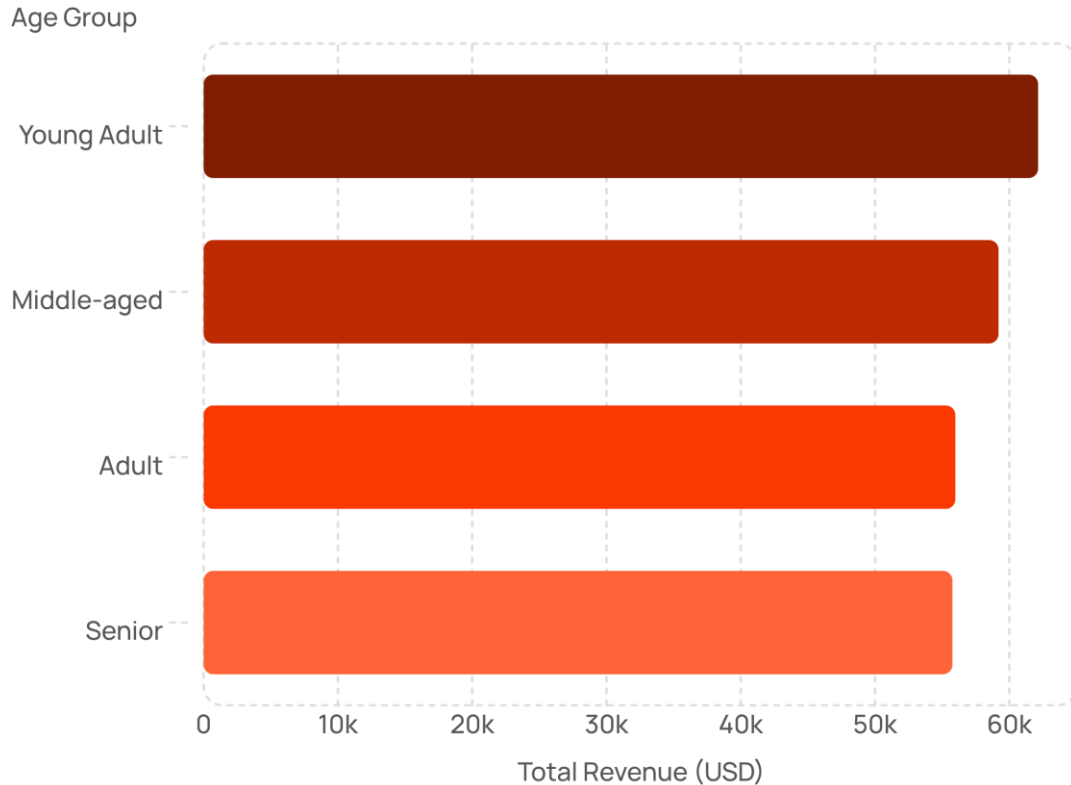
Sandals · Shoes · Sneakers



Outerwear

Jacket · Coat

Revenue by Age Group



Key Insights

- **Young Adults** lead in revenue (\$62,143)
- Revenue is **relatively balanced** across all age groups
- Express shipping averages **\$60.48** vs. Standard **\$58.46**

Power BI Dashboard



3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

27% Subscribers

Filters: Subscription, Gender, Category, Shipping Type · Visuals: Revenue by Category, Sales by Age Group

Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits

→ Loyalty Programs

Reward repeat buyers → Loyal segment

→ Review Discount Policy

Balance sales lift with margin control

→ Product Positioning

Highlight top-rated, best-selling items

→ Targeted Marketing

Focus on high-revenue age groups & express-shipping users